Captain’s Summer Exchange - Ward Melville - Repaired practice edition - Answer key

Division C | Remote Sensing | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

31. (6 points) Authored replacement for source item Question 31: Solve this case and explain your reasoning: A satellite has four nominal acquisition opportunities in a month, but three are cloud-obscured. How many usable optical observations remain?

reasoning: Separate scheduled opportunities from quality-controlled observations.
Reject the three cloud-obscured scenes for surface optical measurement.
Retain one usable scene and report the temporal gap. (3 points).

conclusion: Nominal revisit does not imply four usable measurements. (3 points).

33. (2 points) Authored replacement for source item Question 33: What does a sensor directly record in optical imaging?

Unit: Remote sensing foundations and history.
Worked-case data: An image becomes brighter after a display stretch, but its stored reflectance values are unchanged. Has surface reflectance increased?

A. Tree species with certainty

B. Human land-use intent

C. A complete climate mechanism

D. A band-dependent energy response

Correct: D